higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CONTROL
NQF LEVEL 4

19 NOVEMBER 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 1.1.1 Data derived from measurements on a continuous scale such as time, height and length.

1.1.2 Service or procedure that doesn't fulfil the requirement stated in a contract, drawing or specification.

1.1.3 When standards appropriate for a specific product or service are met.

1.1.4 It is a way of monitoring or measuring a process through the use of statistical techniques.

(4 × 2) (8)

1.2
- Test the quality of the incoming material
- Test the quality of the processed product
- Test the quality of the final product (3)

1.3
- Identify sudden & gradual deviations in quality from the desired value against interference
- Used when inspecting a production line to monitor and manage a controlled production process
- Used to determine whether a process is under control or not - to be used to plan functions, inspections and interventions
- Used to determine or detect any factor that influences the process and disturbance that can be connected with the proper intervention
- To determine whether the intervention was successful or not
- Provides information of the predictability and stability of a process with regard to central tendency and variation (Any 5 × 1) (5)

1.4
- The data may no longer be representative
- Wrong findings may cause resources to be wasted
- Time is wasted to investigate problems that not exist (Any 2 × 1) (2)

1.5 Occupational health and safety (2)

1.6 Orifice plate
Venturi tube
Flow nozzle
Dall tube
Pilot static tube

(Any 5 × 1) (5)
[25]

QUESTION 2

2.1
- Sample withdrawal
- Extractive sampling
- In-situ probing
- Non-invasive testing (Any 3 × 1) (3)

2.2
- Expensive to design
- Not reliable
- High maintanance costs
- Consume a lot of electricity
- Calibration issues
- Time consuming
- Hard to introduce a standard for instrument verification
- Must be monitored continuosly to ensure that the problem is detected in time (Any 2 × 1) (2)

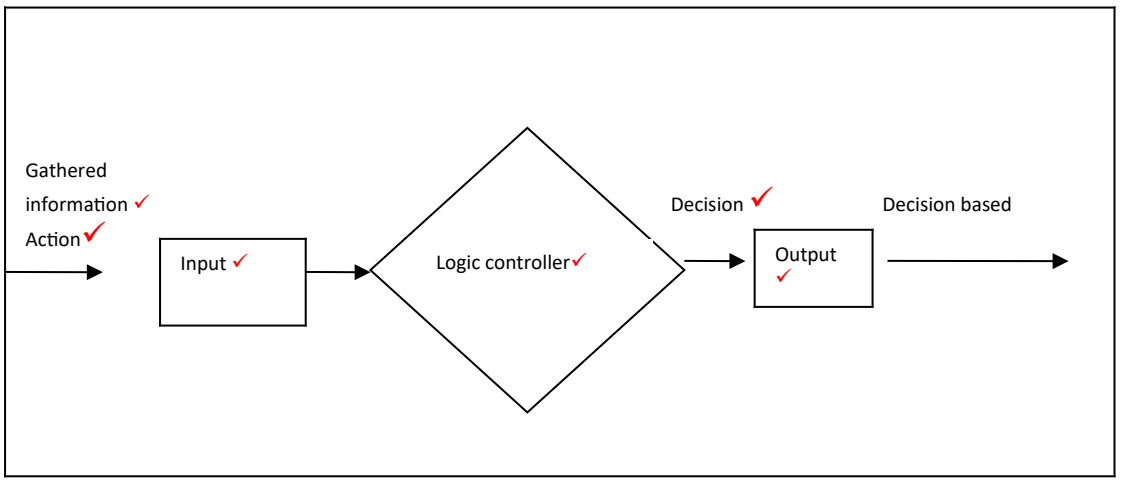
2.3
- Manual
- Automatic
- Cascading (3)

2.4
- Manual control means that the operator sets the controller's out signal directly to the system.
- The set point is not used in this type of control.
- Manual control is used when a process system is not operating normally.
- This usually done during start-up or shutdown.
- When an operator needs to take direct control of a process variable. (5)

2.5
- An emergency
- Planned or breakdown maintenance
- Cleaning, replacement of catalysts or resins
- Equipment change-over
- Plant modification
- Product changes
- Product quality problems
- Raw material shortage
- Excessive production of a product
- Storage capacity limitations (Any 6 × 1) (6)

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2.6



(6)
[25]

QUESTION 3

3.1 It is a system that is used to automatically control processes such as chemical production plants, petroleum refineries and papermaking industries. (2)

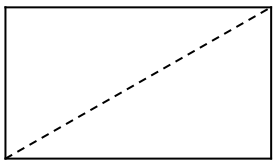
3.2
- Information flow
- Decision making (2)

3.3
- Voltage signal
- Current signal
- Pneumatic signal (3)

3.4
- It is a device that is usually placed at a location far away from the central plant
- Its function is to monitor and make simple decisions
- It slowly relays information to the central unit
- Can also receive instructions (4)

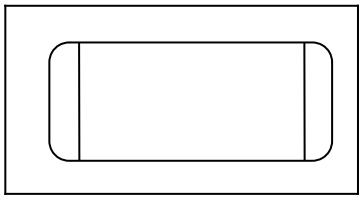
3.5
- To check the settings of a previous batch
- To determine why deviation occurred
- To determine why product quality is not good
- To determine why an accident occurred
- To solve problems that occur in a plant
- For comparison or selection of running conditions
- Trend analysis
- Training
- Best practice determination (Any 6 × 1) (6)

3.6 3.6.1



✓✓✓

3.6.2



✓✓✓

(2 × 3) (6)

3.7
- Spreadsheet
- Manual paper logsheet (2)

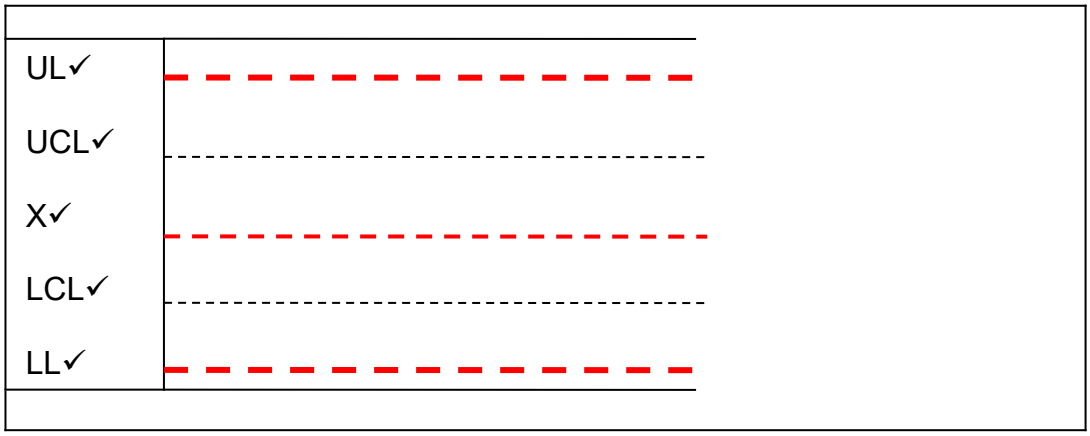
[25]

QUESTION 4

4.1
- For real or short time intervals
- For a long period
- Graphical
- Numerical using data table
- Numerical using comma-separated values (csv)
- For multiple control instruments
- For a specific control instrument (Any 5 × 1) (5)

4.2
- Data retrieval screens are user friendly and easy to work with
- They use screen menus that guide the process controller through the process
- These menus allow one to select different instruments
- The time period is needed
- The format of the data (5)

4.3



(5)

4.4 Abnormal deviations occur because of some specific cause or problem in a system process, which can be addressed and rectified. (2)

4.5 4.5.1 When the quality and properties of raw materials used in a process are changed, that may cause deviations in a process.

4.5.2 Damaged or broken equipment or machines may cause deviations, thus equipment should always be maintained to avoid deviations.
(2 × 2) (4)

4.6
- Control covers that prevent unplanned operation of the control
- Emergency stop controls like panic buttons, bars or pull cords that stop machines in an emergency
- Two-hand controls that require the operator to have both hands on the controls and therefore out of the machine before it can be started
- Presence sensing devices that stop the machine if a worker gets too close (4)

[25]

TOTAL: 100

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